

SOPHIE CAMILLE CLARK

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EDUCATION

University of Arizona 8/2025 - present
Ph.D. in Planetary Sciences
Advisor: Dr. Ilaria Pascucci
Relevant Coursework: Principles of Planetary Physics, Writing Across Space Sciences, Stars & Planets, Cosmochemistry, Astronomical Instrumentation

University of Florida 8/2020 - 12/2024
B.S. in Physics & Astrophysics, *summa cum laude* Cumulative GPA: 3.98/4.0
Thesis Advisor: Dr. Elizabeth Lada

RESEARCH EXPERIENCE

University of Arizona, Tucson, AZ 8/2025 - present
Graduate Research Assistant (Advisor: Dr. Ilaria Pascucci)
Using modeling & JWST observations to characterize MHD outflows from protoplanetary disks

University of Florida, Gainesville, FL 1/2021 - 7/2025
Undergraduate Research Assistant (Advisor: Dr. Elizabeth Lada)
Analyzed Gaia, 2MASS, & Spitzer data to constrain star cluster membership, proper motions, and correlation to large feedback events in Orion

SETI Institute, Mountain View, CA 6/2024 - 8/2024
REU Intern (Advisor: Dr. Uma Gorti)
Developed software to model protoplanetary disk winds and applied it to JWST observations to infer mass loss rates & morphology

California Institute of Technology, Pasadena, CA 6/2023 - 8/2023
Summer Undergraduate Research Fellow (Advisor: Dr. Matthew Graham)
Constrained AGN luminosity variability using an automated machine learning pipeline to analyze TESS and ZTF time series data

PUBLICATIONS

Clark, Gorti, Pascucci, Beck, Bajaj (in prep.) *A low mass loss rate in the inner disk wind of HH30 as measured by JWST*

N. Bajaj, I. Pascucci, S. Cabrit, S. Edwards, G. Cugno, A. Sellek, J. Najita, K. Zhang, R. Alexander, G. Herczeg, U. Gorti, **S. Clark**, & T. Beck (2026), *JWST/MIRI Reveals the Evolution from Molecular to Atomic Disk Winds*, The Astronomical Journal, [arXiv:2607.21733](https://arxiv.org/abs/2607.21733)

HONORS & AWARDS

Best GSC Talk, Lunar & Planetary Laboratory Conference (LPLC) 2026

GRFP Honorable Mention, National Science Foundation (NSF) 2026

Astronomy Department Best Thesis Award, University of Florida 2024

Distinguished Undergraduate Awards Honorable Mention, USRA 2024

SETI Institute REU Award of Excellence, SETI Institute 2024

Chambliss Astronomy Achievement Student Award, American Astronomical Society 2023

University Scholars Program (USP) Research Award, University of Florida 2023

PRESENTATIONS

Modeling JWST Observations of Protoplanetary Disk Outflows with RADMC-3D

1/15/25, The 245th Meeting of the American Astronomical Society, Oral Sessions

A High-Resolution Study of AGN Variability with TESS

1/10/24, The 243rd Meeting of the American Astronomical Society, Oral Sessions

Analyzing the Structure and Motions of the NGC 1977 Cluster with Gaia DR3

6/6/23, The 242nd Meeting of the American Astronomical Society, iPoster Sessions

TEACHING EXPERIENCE

Post-baccalaureate Teaching Assistant, University of Florida

PHY2049L, Physics 2 Laboratory (3 sections)

Spring 2025

IDS2935, The Art & Science of Astrophotography

Spring 2025

Undergraduate Teaching Assistant, University of Florida

IDS2935, The Art & Science of Astrophotography

Fall 2023, Spring 2024, Fall 2024

MAC2312, Analytic Geometry & Calculus 2

Spring 2023

HUM2930, Research & Creativity

Summer 2021, Fall 2021

LEADERSHIP & OUTREACH

Outreach Volunteer, University of Florida Department of Astronomy

2022 - 2025

- Various events, usually operating telescopes for the public to observe

URSP Student Advisory Board, UF Center for Undergraduate Research

2021 - 2024

- External Director, 2023 - 2024

- Director of Research Seminars, 2022 - 2023

- Research Seminar Committee Member, 2021 - 2022

Mentor, Women in Astronomy & Astrophysics Mentorship Program (WAAM)

2023 - 2024

- Mentored undergraduate students on summer research opportunities and career plans

PROGRAMMING SKILLS

Languages: Python, L^AT_EX, MATLAB

Systems: Linux/Unix, Windows OS, Mac OS, Git

Applications: data analysis, image processing, MCMC, machine learning

Facilities: Gaia, 2MASS, Spitzer, TESS, JWST, ALMA

MEDIA APPEARANCES

[SETI Institute to Honor Contributions to Exoplanet Research with its Carl Sagan Center Director's Award](#), SETI Institute Press Release

[A Picture is Worth a Thousand Stars](#), Florida Gator Magazine Fall 2024

[Stars in Astronomy visible right here on campus](#), UF Liberal Arts & Sciences News